This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

UTPTrack: Towards Simple and Unified Token Pruning for Visual Tracking

Hao Wu[1,2*] Xudong Wang[1*] Jialiang Zhang[1] Junlong Tong[1,2,3]
Xinghao Chen[1,4] Junyan Lin[1,4] Yunpu Ma[5] Xiaoyu Shen[1,2†]
[1]Institute of Digital Twin, Eastern Institute of Technology, Ningbo
[2]Ningbo Key Laboratory of Spatial Intelligence and Digital Derivative
[3]Shanghai Jiao Tong University [4]The Hong Kong Polytechnic University
[5]Munich Center for Machine Learning, LMU Munich
haowu.ai.research@gmail.com xyshen@eitech.edu.cn

Abstract

One-stream Transformer-based trackers achieve advanced performance in visual object tracking but suffer from significant computational overhead that hinders real-time deployment. While token pruning offers a path to efficiency, existing methods are fragmented. They typically prune the search region, dynamic template, and static template in isolation, overlooking critical inter-component dependencies, which yields suboptimal pruning and degraded accuracy. To address this, we introduce ***UTPTrack****, a simple and Unified Token Pruning framework that, for the first time, jointly compresses all three components. UTPTrack employs an attention-guided, token type-aware strategy to holistically model redundancy, a design that seamlessly supports unified tracking across multimodal and language-guided tasks within a single model. Extensive evaluations on 10 benchmarks demonstrate that UTPTrack achieves a new state-of-the-art in the accuracy-efficiency trade-off for pruning-based trackers, pruning 65.4% of vision tokens in RGB-based tracking and 67.5% in unified tracking while preserving 99.7% and 100.5% of baseline performance, respectively. This strong performance across both RGB and multimodal scenarios underlines its potential as a robust foundation for future research in efficient visual tracking. Code will be released at https://github.com/EIT-NLP/UTPTrack.*

1. Introduction

Visual Object Tracking (VOT) aims to estimate the position and shape of a target throughout a video sequence given its initial state [3, 4, 45]. A typical tracker represents the target with a *static template (ST)* from the first frame, optionally updates a *dynamic template (DT)* to handle appearance variations, and localizes it within a cropped *search region (SR)* in subsequent frames [37, 43]. Owing to its dynamic nature, VOT requires strong spatio-temporal modeling [9, 16, 23]. Recent advances in Transformer-based architectures have shown strong modeling capabilities and brought significant improvements in tracking performance [4, 25, 38, 43].

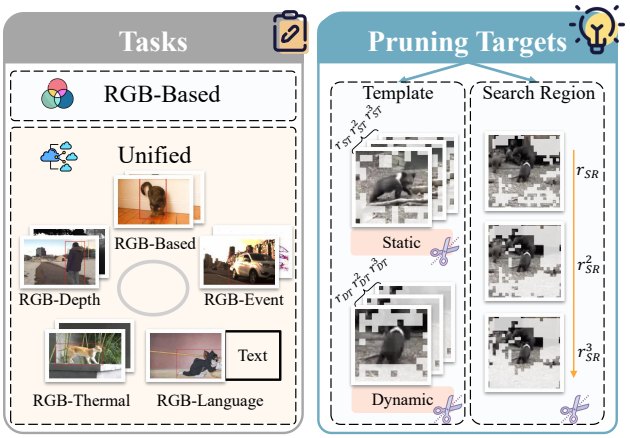


Figure 1. UTPTrack supports RGB-based and unified tracking, prunes redundant tokens in the search region (SR), dynamic template (DT), and static template (ST) to improve efficiency, with r indicating the token retention ratio for each component.

Transformer-based trackers generally categorized as two-stream or one-stream. Two-stream trackers [4, 5, 48] process the template and search region separately, allowing template reuse and reducing inference cost. In contrast, one-stream trackers [2, 7, 41, 45] jointly encode both inputs within a unified Transformer, enabling richer template–search interactions and stronger global feature representations. While one-stream trackers have become the dominant paradigm due to their superior performance, they come with significant computational overhead: the quadratic complexity of Transformers, compounded by the

*Equal contribution.
†Corresponding author.

large number of video tokens, makes real-time deployment particularly challenging on resource-constrained devices.

Token pruning has recently improved tracking efficiency by reducing tokens in the search region or the dynamic template [19, 45, 46]. However, current methods underexploit their potential since none performs joint pruning across all three critical components: SR, DT, and ST. Treating these components in isolation overlooks their inherent interdependencies, which are essential for precise localization and boundary-aware tracking. Without a unified redundancy modeling strategy that captures cross-component relationships, existing approaches often make suboptimal decisions: informative tokens may be discarded, and uneven redundancy across components remains underexploited. Consequently, the tracking performance is often compromised through degraded spatial consistency and semantic completeness, a problem that becomes particularly severe in multimodal scenarios where aligning information is critical.

To address this problem, we propose *UTPTrack*, a simple and Unified Token Pruning framework for efficient and accurate tracking (Fig. 1). Unlike previous methods that prune tokens from a single component or rely on heuristic rules, *UTPTrack* jointly prunes all three sources (search region, dynamic template, and static template) within a one-stream architecture, guided by a holistic redundancy modeling strategy. Specifically, we identify redundancy patterns within each component and tailor attention-guided pruning strategies accordingly. For the search region and dynamic template, token relevance is computed based on similarity to the static template center token, retaining only the most important tokens. For the static template, we further enhance pruning robustness via a token type-aware strategy, leveraging spatial priors from the target bounding box to avoid discarding foreground tokens.

Beyond RGB-based tracking, UTPTrack naturally extends to unified tracking, which supports multimodal and language-guided tasks in one model. For additional visual modalities (*e.g.*, depth, thermal, event), pruning is performed in a shared embedding space using the same attention-based mechanism. For RGB–language tasks, we introduce a text-guided pruning strategy, where semantic cues from language tokens jointly guide token selection alongside visual features. This unified and modality-aware design allows UTPTrack to generalize across diverse tracking scenarios while maintaining high efficiency.

We comprehensively evaluate the effectiveness of UTPTrack on OSTrack [45] and SUTrack [6] across ten RGB-based and multimodal tracking benchmarks. Extensive results demonstrate its strong generalization and efficiency across diverse tasks. Specifically, UTPTrack reduces token number by 65.4% on OSTrack_{384} and 67.5% on SUTrack_{384}, with corresponding MAC reductions of 31.3% and 28.4%, while maintaining or even slightly improving accuracy (99.7% and 100.5% of baseline performance). The contributions of this work are summarized as follows:

- We introduce *UTPTrack*, the first unified token pruning framework that jointly compresses the SR, DT, and ST within a one-stream Transformer.
- We propose an *attention-guided, token type-aware strategy* using cross-component similarity and spatial priors to remove redundancy while preserving key information.
- We extend our framework to *unified tracking* via a unified, modality-aware pruning mechanism, including a novel text-guided strategy that integrates semantic cues from natural language.
- Extensive experiments across ten benchmarks demonstrate that UTPTrack significantly reduces token count and computation while maintaining state-of-the-art performance in both RGB-based and unified tracking.

2. Related Work

RGB-based and Unified Object Tracking. RGB-based object tracking focuses on a single target with RGB images and underpins many advances. Recently, one-stream Transformer trackers couple feature extraction and relation modeling for stronger representations and more accurate localization [2, 7, 12, 45]. Building on this flexibility, research has moved to unified frameworks handling multiple modalities (RGB+Depth/Thermal/Event/Language). Parameter-efficient adaptation [13, 14, 47] keeps a strong RGB foundation while injecting modality cues via lightweight prompts or adapters. UnTrack [40] learns a shared low-rank latent space across modalities, whereas SUTrack [6] consolidates five tasks within a single one-stream model via a unified modality representation. For efficiency, some work reduces redundancy through shared backbones, compact experts, distillation [28], or learnable interaction tokens [15]. Complementary to these directions, we address the remaining token bottleneck with a token-level economy using hierarchical pruning, improving the compute–performance trade-off without modifying unified architectures.

Token Compression in Computer Vision. Token compression in ViT comprise two families: pruning and merging [29, 39]. **Pruning** removes low-importance tokens [11]. DynamicViT [34] predicts token saliency with an MLP and discards the rest. EViT [24] and SPViT [17] estimate importance and either drop or collapse low-saliency tokens before the next layer. Evo-ViT [42] uses evolving CLS attention with slow–fast updates to preserve structure. **Merging** combines similar tokens. ToMe [1] performs bipartite soft matching to merge a fixed quota of similar tokens. VoMix [32] redistributes redundant token content to key tokens via similarity-weighted mixing. ATM [20] adopts a layer-wise decreasing similarity threshold and a merging-aware matching strategy. Despite extensive study

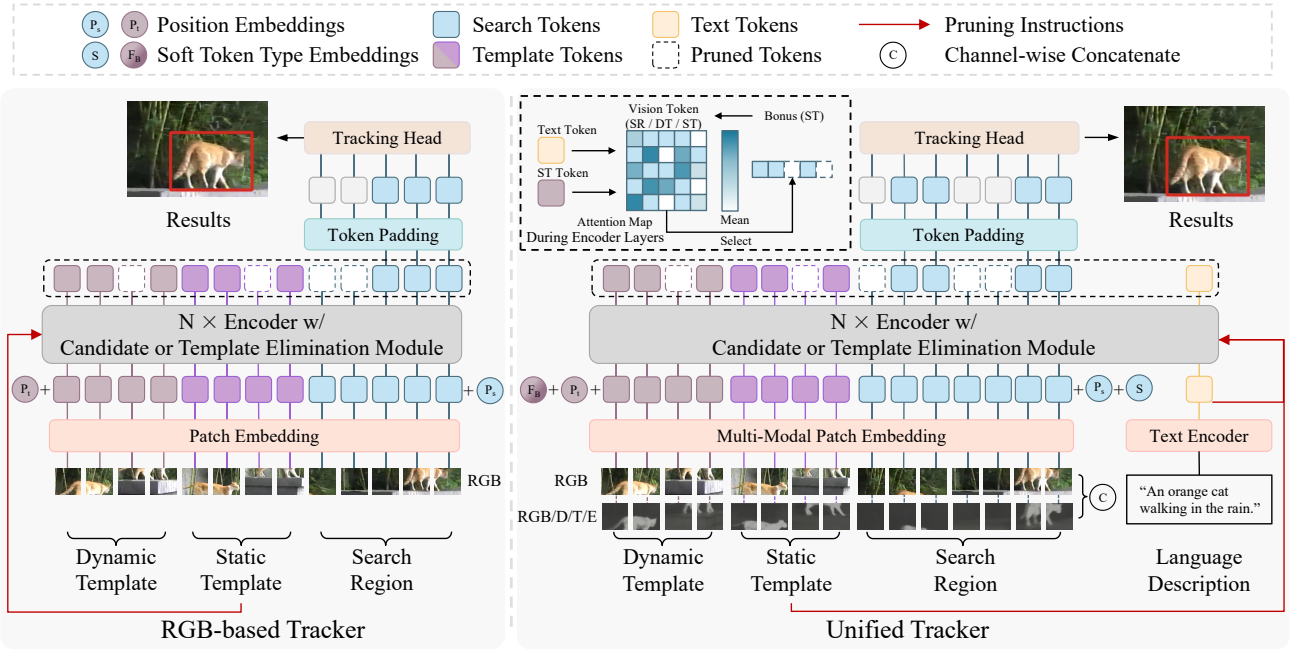


Figure 2. Architecture of the proposed UTPTrack. UTPTrack supports both RGB-based and unified tracking. It adopts a one-stream transformer that jointly processes tokens from the search region (SR), dynamic template (DT), and static template (ST). A lightweight Candidate or Template Elimination Module (CTEM) is inserted into encoder layers to prune redundant tokens from all three sources. In the figure, D/T/E denote depth, thermal, and event modalities, respectively.

in ViTs, token compression for tracking remains underexplored. OSTrack [45] proposes early elimination of search tokens based on similarity to the central static template. ProContEXT [19] refines this by incorporating both static and dynamic templates to compute similarity scores to improve robustness. However, these methods prune components in isolation, overlooking redundancy overlap and cross-dependencies among the search region, dynamic template, and static template. In contrast, our method introduces a joint pruning strategy across all three components, aiming to model redundancy holistically and improve pruning effectiveness in more complex tracking scenarios.

3. Methodology

In this section, we present UTPTrack. We first describe the one-stream pipelines for RGB-based and unified tracking, together with token-pruning preliminaries, then analyze pruning targets (*i.e.*, SR, DT, ST). We further introduce a token type-aware strategy for ST and extend the framework to unified tracking with a modality-aware pruning scheme.

3.1. Preliminary

RGB-based Tracking Pipeline. Given a ST $\mathbf{Z_s} \in \mathbb{R}^{H_z \times W_z \times 3}$ and the current frame's SR $\mathbf{X} \in \mathbb{R}^{H_x \times W_x \times 3}$, the tracker aims to estimate the target bounding box $\mathbf{B} \in \mathbb{R}^4$. To handle appearance changes, a DT $\mathbf{Z_d} \in \mathbb{R}^{H_z \times W_z \times 3}$ is often maintained. A typical one-stream RGB tracker uses a backbone $\mathcal{F}(\cdot)$ for feature extraction and relation modeling, followed by a tracking head $\varphi(\cdot)$. The inputs are split into patches, embedded, and flattened into token sequences: $\mathbf{E_x} \in \mathbb{R}^{N_x \times D}$, $\mathbf{E_{sz}} \in \mathbb{R}^{N_{sz} \times D}$, and $\mathbf{E_{dz}} \in \mathbb{R}^{N_{dz} \times D}$. These are concatenated and processed by the backbone: $\mathbf{B} = \varphi\left(\mathcal{F}(\text{Concat}(\mathbf{E_x}, \mathbf{E_{sz}}, \mathbf{E_{dz}}))\right)$.

Unified Tracking Pipeline. Unified trackers (*e.g.*, SUTrack) extend RGB trackers by integrating complementary modalities (*e.g.*, depth, thermal, event, language) into a single framework. For depth, thermal, and event, three additional channels (D/T/E) are concatenated with RGB to form a six-channel input. The multimodal features are aligned and projected into unified embeddings, yielding SR $\mathbf{E_x} \in \mathbb{R}^{N_x \times 2D}$, ST $\mathbf{E_{sz}} \in \mathbb{R}^{N_{sz} \times 2D}$ and DT $\mathbf{E_{dz}} \in \mathbb{R}^{N_{dz} \times 2D}$. For language, a single token is extracted via CLIP-L [33] from the text description and concatenated with visual tokens, enabling cross-modal interaction. The output is computed as: $\mathbf{B} = \varphi\left(\mathcal{F}(\text{Concat}(\mathbf{E_x}, \mathbf{E_{sz}}, \mathbf{E_{dz}}, \mathbf{E_{text}}))\right)$. Missing modalities are handled by duplicating RGB channels (for D/T/E) or using a fixed dummy sentence (for text), ensuring consistent input format.

As shown in Fig. 2, both pipelines share a similar structure, with the unified version extending the input for multimodal data while remaining compatible with RGB.

Token Pruning in Tracking. In ViT [8], MHA and MLP process all tokens, incurring $\mathcal{O}(N^2)$ and $\mathcal{O}(N)$ costs, which hinders real-time tracking. Token pruning reduces compute

by keeping salient tokens, but tracking requires spatial coherence. Therefore, we restore retained tokens to their original indices and zero-pad pruned slots before the tracking head, preserving spatial layout while reducing computation.

3.2. UTPTrack

The architecture of UTPTrack is shown in Fig. 2. Based on the RGB and unified pipelines (Sec. 3.1), it adopts a one-stream design that jointly processes SR, ST, and DT tokens for tracking. Following OSTrack [45], we insert a lightweight Candidate or Template Elimination Module (CTEM) into selected layers to prune redundant tokens using attention-derived importance scores. Pruning is guided by the ST in RGB tracking, and additionally by language in unified tracking. To maintain spatial alignment, pruned SR tokens are zero-padded to their original positions before the tracking head. Details of CTEM are provided in Sec. 3.2.1.

3.2.1. Token Pruning for RGB-based Tracking

In the one-stream architecture, SR, ST, and DT tokens are jointly processed via attention layers. While this enables rich cross-token interactions, it also introduces redundancy, especially from background regions with limited discriminative value. To mitigate this, we design dedicated elimination modules to prune unimportant tokens from each source.

Pruning is guided by attention-based similarity, reusing attention weights from the transformer encoder to measure token relevance without extra computation. Given the concatenated token sequence $[\mathbf{E_x}; \mathbf{E_{sz}}; \mathbf{E_{dz}}]$ from $\mathbf{X}$, $\mathbf{Z_s}$, and $\mathbf{Z_d}$, the attention is computed as:

$$\begin{aligned}\text{Attention}(Q,K,V) &= \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \\ &= \text{Softmax}\left(\frac{[Q_x;Q_{sz};Q_{dz}][K_x;K_{sz};K_{dz}]^T}{\sqrt{d_k}}\right)\begin{bmatrix}V_x\\V_{sz}\\V_{dz}\end{bmatrix}\end{aligned} \tag{1}$$

where Q, K, and V denote query, key, and value matrices, with subscripts x, sz, dz indicating the SR, ST, and DT, respectively. The attention weights A in Eq. (1) expand to:

$$A = \text{Softmax}\left(\frac{1}{\sqrt{d_k}}\begin{bmatrix}Q_xK_x^T & Q_xK_{sz}^T & Q_xK_{dz}^T\\ Q_{sz}K_x^T & Q_{sz}K_{sz}^T & Q_{sz}K_{dz}^T\\ Q_{dz}K_x^T & Q_{dz}K_{sz}^T & Q_{dz}K_{dz}^T\end{bmatrix}\right) \tag{2}$$

This dense interaction benefits feature fusion but risks noise propagation. For instance, terms like $Q_xK_x^T$, $Q_xK_{sz}^T$, and $Q_xK_{dz}^T$ may allow background tokens to attend to each other or dominate target-related responses. Similarly, template tokens attending to noisy search tokens (*e.g.*, via $Q_{sz}K_x^T$, $Q_{dz}K_x^T$) can degrade representation quality.

Candidate (Search Region) Elimination. The SR tokens often contain background clutter, introducing redundancy and harming localization accuracy. This is evident in attention terms like $Q_xK_x^T$, where background tokens attend to each other, and $Q_xK_{sz}^T$ or $Q_xK_{dz}^T$, where SR tokens attend to non-discriminative template regions. To address this, we measure the importance of each SR token based on its attention similarity to the ST. According to Eq. (2), the relevance score is computed by $\omega_x = \text{softmax}(Q_{sz'}K_x^T/\sqrt{d_k}) \in \mathbb{R}^{1\times N_x}$, where $Q_{sz'}$ is the query of the ST's center token and N_x is the number of SR tokens. We retain the top-k tokens and prune the rest, reducing background interference while preserving target-relevant information.

Dynamic Template Elimination. The DT may contain noisy tokens due to drift, occlusion, or appearance changes, which can negatively affect the SR and ST via attention terms like $Q_{dz}K_x^T$ and $Q_{dz}K_{sz}^T$. Similar to the SR, we compute the similarity between each DT token and the ST's center token as $\omega_{dz} = \text{softmax}(Q_{sz'}K_{dz}^T/\sqrt{d_k}) \in \mathbb{R}^{1\times N_{dz}}$, and pruning those with low relevance.

Static Template Elimination. Since the ST is obtained by enlarging the target box, it may include background tokens that introduce irrelevant attention interactions (e.g., $Q_{sz}K_x^T$ and $Q_{sz}K_{dz}^T$). To refine its representation, we measure each token's similarity to the center token as $\omega_{sz} = \text{softmax}(Q_{sz'}K_{sz}^T/\sqrt{d_k}) \in \mathbb{R}^{1\times N_{sz}}$, and prune tokens with low relevance, while always preserving the center token.

Token Type Aware Pruning. Motivated by the token-type embedding mechanisms [6, 26], we propose a token type-aware pruning strategy during ST elimination to suppress the probability of mistakenly discarding foreground tokens. Given a ST $\mathbf{Z_s} \in \mathbb{R}^{H_z\times W_z\times 3}$ and its bounding box $\mathbf{B}$, we construct a binary mask $\mathbf{M} \in \mathbb{R}^{H_z\times W_z}$, where pixels inside $\mathbf{B}$ are set to 1 and those outside to 0:

$$M(i,j) = 1 \text{ if } (i,j) \text{ is inside } B,\ 0 \text{ otherwise.} \tag{3}$$

We divide the mask into non-overlapping P^2 patches $\mathbf{M}_{\text{patch}}^{(k)}$ and assign each patch a foreground score by averaging its mask values. This score serves as a bonus to encourage retaining foreground tokens during pruning and is incorporated into the attention-based elimination module (see Fig. 2). We explore three bonus strategies to control how this foreground prior influences pruning.

- Full bonus assigns 1 if all pixels lie inside the bounding box. The value for patch $\mathbf{M}_{\text{patch}}^{(k)}$ is:

$$b_{\text{full}}^{(k)} = 1 \text{if } \mathbf{M}(i,j) = 1 \text{ for all } (i,j) \in \mathbf{M}_{\text{patch}}^{(k)},\ 0 \text{ otherwise.} \tag{4}$$

- Soft bonus uses the mean mask value of each patch:

$$b_{\text{soft}}^{(k)} = m_{\text{avg}}^{(k)} = \frac{1}{P^2}\sum \mathbf{M}(i,j), \quad (i,j) \in \mathbf{M}_{\text{patch}}^{(k)}. \tag{5}$$

- All bonus assigns 1 if any pixel lies inside the box:

$$b_{\text{all}}^{(k)} = 1 \text{ if } \exists (i,j) \in \mathbf{M}_{\text{patch}}^{(k)} : \mathbf{M}(i,j) = 1,\ 0 \text{ otherwise.} \tag{6}$$

These bonuses are added directly to attention scores during ranking to guide pruning toward foreground-relevant tokens. As shown in Fig. 2, this mechanism is integrated into the elimination module. The soft version is used by default.

3.2.2. Token Pruning for Unified Tracking

The proposed pruning strategy, originally designed for RGB tracking, naturally extends to multimodal settings. For visual modalities (RGBD, RGBT, RGBE), auxiliary channels (depth, thermal, event) are concatenated with RGB to form a six-channel input. After projection into a unified embedding space, token dimensions increase, but their spatial layout remains unchanged. Thus, pruning based on attention to the ST's center token remains effective without modification. For language modality, we additionally account for semantic cues: the text description is encoded into a single token using CLIP-L [33] and concatenated with visual tokens before the transformer, extending the attention in Eq. (2) to include interactions with the language token. The updated attention weights A are:

$$A{=}\text{Softmax}\left(\frac{1}{\sqrt{d_k}}\begin{bmatrix} Q_x K_x^T & Q_x K_{sz}^T & Q_x K_{dz}^T & Q_x K_t^T \\ Q_{sz} K_x^T & Q_{sz} K_{sz}^T & Q_{sz} K_{dz}^T & Q_{sz} K_t^T \\ Q_{dz} K_x^T & Q_{dz} K_{sz}^T & Q_{dz} K_{dz}^T & Q_{dz} K_t^T \\ Q_t K_x^T & Q_t K_{sz}^T & Q_t K_{dz}^T & Q_t K_t^T \end{bmatrix}\right) \quad (7)$$

Text-Guided Pruning. Based on the updated attention in Eq. (7), the language token interacts with all visual tokens through bidirectional attention, enabling semantic cues to enhance spatial representations. To leverage this for pruning, we jointly use the ST's center token and the language token to guide token importance estimation. Specifically, the importance of each token is computed as:

$$\omega_x = \phi\left(\text{softmax}\left(\frac{Q_{sz'} K_x^T}{\sqrt{d_k}}\right) + \text{softmax}\left(\frac{Q_t K_x^T}{\sqrt{d_k}}\right)\right) \quad (8)$$

where $\phi(\cdot)$ denotes summation across attention maps, and Q_t is the query of the text token. This enables pruning to benefit from both spatial and semantic guidance. The same principle can applie to the DT and ST. Related ablations are provided in Sec. 4.3, where we study which token group benefit most from text-guided pruning.

3.3. Training and Inference

Following OSTrack [45], our RGB-based trackers use a weighted focal loss for classification, L1 and generalized IoU losses for bounding box regression. The total loss is:

$$\mathcal{L}_{\text{RGB}} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{giou}}\mathcal{L}_{\text{giou}} + \lambda_{L_1}\mathcal{L}_{L_1}, \quad (9)$$

where $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{giou}}$, and $\mathcal{L}_{L_1}$ denote the weighted focal, generalized IoU, and L_1 losses. We set $\lambda_{\text{cls}} = 1$, $\lambda_{\text{giou}} = 2$, and $\lambda_{L_1} = 5$. For unified tracking, following SUTrack [6], we incorporate a cross-entropy loss for task recognition:

$$\mathcal{L}_{\text{Unified}} = \mathcal{L}_{\text{RGB}} + \lambda_{\text{task}}\mathcal{L}_{\text{task}}, \quad (10)$$

where $\mathcal{L}_{\text{task}}$ is the task-recognition cross-entropy loss, with $\lambda_{\text{task}} = 1$. During inference, RGB-based and unified trackers adopt a dual-template strategy using one ST and one DT. The DT is updated based on a fixed time interval and confidence threshold, following the strategy of STARK [43].

Table 1. Details of UTPTrack model variants.

Method	Base Model	Params (M)	MACs (G)	FPS (GPU)	FPS (CPU)
UTPTrack-O_{256}	OSTrack-256	92	24 (-11)	95 (+1)	17 (+8)
UTPTrack-O_{384}	OSTrack-384	92	53 (-25)	47 (+7)	6 (+3)
UTPTrack-S_{224}	SUTrack-B224	70 (+85)	16 (-7)	43 (+2)	9 (+1)
UTPTrack-S_{384}	SUTrack-B384	70 (+85)	48 (-19)	31 (+4)	5 (+2)

4. Experiment

4.1. Implementation Details

Model. We present four UTPTrack variants across architectures and resolutions for RGB and unified tracking. Model names encode the search region resolution (number) and base tracker (suffix). OSTrack [45] is a modified version without the candidate elimination module and with a single DT. Table 1 reports parameters, MACs, and inference speed. The negative MAC subscript indicates computation reduction and the FPS subscript denotes the speed gain from pruning, while the parameter subscript marks the CLIP-L [33] text encoder used in language-guided tracking.

Training. Following standard practice in RGB-based tracking [45], UTPTrack-O are trained on TrackingNet [30], LaSOT [9], GOT-10k [16], and COCO [27] for 300 epochs with 60k image pairs per epoch. For unified tracking, UTPTrack-S follows the SUTrack and uses a broader multimodal datasets, including TrackingNet [30], LaSOT [9], GOT-10k [16], COCO [27], TNL2K [35], VASTTrack [31], DepthTrack [44], LasHeR [22], and VisEvent [36], trained for 180 epochs with 100k image pairs per epoch. For all variants, template and search region crops are generated by expanding the target bounding box by $2\times$ and $4\times$, respectively. Training is conducted on 4 NVIDIA A100 GPUs.

Inference. During inference, the DT is updated every 25 frames using a confidence threshold of 0.7. A Hanning window penalty is applied to incorporate position priors, following common practice [6, 45]. Inference speed is evaluated on a single NVIDIA 1080 Ti GPU and an Intel Xeon Gold 6226R @ 2.90GHz CPU.

4.2. State-of-the-Art Comparisons

We verify the effectiveness of UTPTrack under two tracking paradigms: (1) RGB-based tracking and (2) Unified tracking, which includes RGB-based, RGB-Depth, RGB-Thermal, RGB-Event, and RGB-Language tasks. For each paradigm, we use a representative base model (OSTrack or SUTrack) and compare UTPTrack against state-of-the-art token compression methods, including CE [45], ToMe [1], EViT [24], and DynamicViT [34], under varying input resolutions. We report results under two complementary protocols: (1) Overall Performance, using released default settings, and (2) Controlled-Budget Performance, evaluated at three matched token compression ratio.

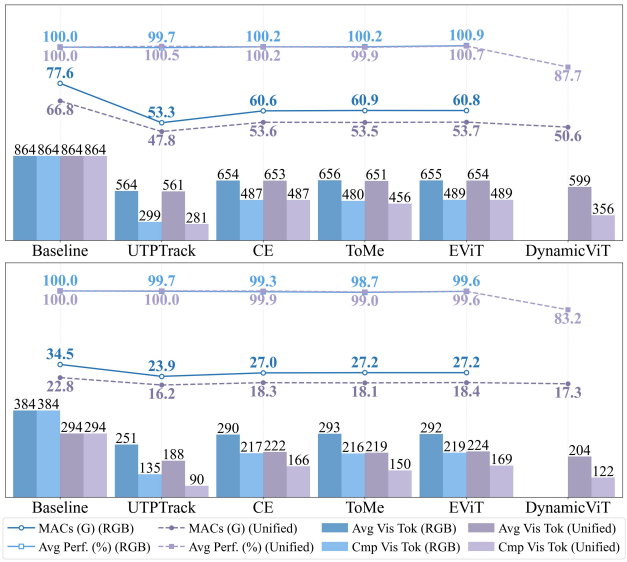


Figure 3. Performance comparison of UTPTrack and other pruning methods under each method's default compression settings at two resolutions. Top (High Resolution): 384 (RGB and Unified). Bottom (Low Resolution): 256 (RGB), and 224 (Unified).

Overall Performance. As shown in Fig. 3, UTPTrack achieves the best performance-efficiency trade-off across both RGB-based and Unified tracking settings. At the low-resolution setting, it attains the highest compression ratio while delivering the best performance in both paradigms. Specifically (RGB-based / Unified), MACs drop by 30.7% (34.5G $\rightarrow$ 23.9G) / 28.9% (22.8G $\rightarrow$ 16.2G) and vision tokens prune by 64.8% (384 $\rightarrow$ 135) / 69.4% (294 $\rightarrow$ 90), while maintaining 99.7% / 100.0% of baseline performance. Under comparable performance, it further achieved a 1.5$\times$ / 1.6$\times$ compression. At the high-resolution setting, UTPTrack sustains its advantage and scales favorably: (RGB-based / Unified) MACs are reduced by 31.3% / 28.4%, tokens by 65.4% / 67.5%, with performance preserved at 99.7% / 100.5% of the baseline. Notably, as the resolution increases, UTPTrack-S even surpasses the baseline, reaching 100.5% performance. This small but consistent gain suggests that our token selection acts as a mild regularizer by removing redundant or noisy visual tokens and sharpening attention to salient regions, thereby improving accuracy while reducing computation. Comprehensive per-benchmark and per-metric results are provided in Appendix C, with detailed practical efficiency analyses in Appendix B. Unlike methods such as ToMe and DynamicViT that rely on external heuristics or auxiliary prediction modules, UTPTrack directly utilizes the model's inherent attention maps to guide token pruning. This internal mechanism preserves alignment with task-specific semantics while introducing minimal architectural overhead. Mover, it is lightweight and architecture-agnostic, making it compatible with any transformer-based tracker without the need for retraining or structural modifications.

Table 2. Performance comparisions under different vision token configurations across **RGB-based tracking**. All methods are applied on the same base model **OSTrack$_{256}$**. The average performance listed is calculated across all four benchmarks.

Method	*LaSOT*	*LaSOT$_{ext}$*	*TrkNet*	*GOT-10k*	Avg Perf. (%)
Upper Bound, 384 Tokens (100%)					
OSTrack$_{256}$ (Baseline)	67.9	47.6	84.5	74.9	100%
Retain 87.2% Tokens in Average ($\approx$ 335 tokens, $\downarrow$ 18.8%)					
OSTrack-CE$_{256}$	67.1	46.9	**84.1**	74.9	**99.3%** ($\downarrow$ 0.7%)
OSTrack-ToMe$_{256}$	**68.0**	46.4	83.8	74.3	99.0% ($\downarrow$ 1.0%)
OSTrack-EViT$_{256}$	66.8	46.7	**84.1**	75.0	99.0% ($\downarrow$ 1.0%)
UTPTrack-O$_{256}$ (ours)	67.1	**48.3**	84.0	**75.6**	**100.2%** ($\uparrow$ 0.2%)
Retain 75.5% Tokens in Average ($\approx$ 290 tokens, $\downarrow$ 24.5%)					
OSTrack-CE$_{256}$	67.3	**47.3**	**84.2**	74.2	99.3% ($\downarrow$ 0.7%)
OSTrack-ToMe$_{256}$	**68.2**	46.5	**84.2**	**74.9**	99.4% ($\downarrow$ 0.6%)
OSTrack-EViT$_{256}$	67.8	47.1	83.8	72.9	98.8% ($\downarrow$ 1.2%)
UTPTrack-O$_{256}$ (ours)	67.6	**47.3**	83.8	74.8	**99.5%** ($\downarrow$ 0.5%)
Retain 65.6% Tokens in Average ($\approx$ 252 tokens, $\downarrow$ 34.4%)					
OSTrack-CE$_{256}$	67.2	**46.5**	83.3	72.8	98.1% ($\downarrow$ 1.9%)
OSTrack-ToMe$_{256}$	66.3	46.0	82.5	71.0	96.7% ($\downarrow$ 3.3%)
OSTrack-EViT$_{256}$	68.1	46.1	83.0	72.8	98.1% ($\downarrow$ 1.9%)
UTPTrack-O$_{256}$ (ours)	**68.2**	46.2	**84.0**	**74.9**	**99.2%** ($\downarrow$ 0.8%)

Controlled-Budget Performance. To ensure a fair comparison across methods, we conduct controlled-budget experiment under three different compression ratio at low-resolution setting. For RGB-based tracking, as shown in Tab. 2, we compare UTPTrack-O against other methods across four widely used RGB-based benchmarks. UTPTrack-O consistently outperforms all counterparts at all pruning ratios. Notably, at a moderate budget, UTPTrack-O$_{256}$ surpasses the unpruned baseline even after pruning 18.8% of vision tokens, indicating that a substantial portion of tokens is redundant and that selective pruning can improve generalization. For Unified tracking, as shown in Tab. 3, we compare UTPTrack-S against other methods across elevent widely used RGB-based and multimodal benchmarks. UTPTrack-S$_{224}$ likewise maintains a clear lead at every pruning ratios, confirming that the proposed pruning policy transfers effectively beyond RGB-only settings. Compared with the most similar attention-based token pruning approach, CE [45], UTPTrack achieves higher performance on nearly all benchmarks under the all three pruning ratios for both RGB-based and Unified Tracking. Notably, as the compression ratio increases, the performance gap widens in favor of UTPTrack. We attribute this to our cross-component joint pruning strategy, which effectively identifies and reduces redundancy across all components, better preserving critical areas. The complete per-metrics of each benchmark are provided in Appendix D.

Table 3. Performance comparisons under different vision token compression configurations across **unified tracking**. All methods are applied on the same base model **SUTrack$_{224}$**. The best result are **bolded** and the second best results are underlined in all following tables. The average performance listed is calculated across all 10 benchmarks. TrackingNet is abbreviated as TrkNet for brevity.

Method	RGB				RGB-D	RGB-T		RGB-E	RGB-Lang		Avg Perf. (%)
	LaSOT	*LaSOT$_{ext}$*	*TrkNet*	*GOT-10k*	*VOT-RGBD22*	*LasHeR*	*RGBT234*	*VisEvent*	*TNL2K*	*OTB99*	
Upper Bound, 294 Tokens (100%)											
SUTrack$_{224}$ (Baseline)	73.7	53.2	85.9	77.5	75.5	59.9	70.0	63.3	67.8	68.4	100%
Retain 71.4% Tokens in Average (≈ 218 tokens, ↓ 25.6%)											
SUTrack-CE$_{224}$	**72.4**	52.8	85.1	77.4	75.8	59.9	70.0	61.8	66.0	70.5	99.5% (↓ 0.5%)
SUTrack-ToMe$_{224}$	72.1	52.6	**85.4**	76.9	76.0	60.0	69.0	**62.3**	66.2	70.4	99.4% (↓ 0.6%)
SUTrack-EViT$_{224}$	71.6	52.3	84.9	76.9	75.7	59.0	70.0	62.0	65.7	70.7	99.0% (↓ 1.0%)
SUTrack-DyViT$_{224}$	68.8	49.7	80.2	74.6	74.4	57.6	68.7	61.3	64.0	**71.5**	96.5% (↓ 3.5%)
UTPTrack-S$_{224}$ (ours)	72.1	**52.9**	85.1	**77.7**	**76.5**	**60.1**	**70.2**	62.0	**66.4**	70.7	**99.8%** (↓ 0.2%)
Retain 52.0% Tokens in Average (≈ 153 tokens, ↓ 48.0%)											
SUTrack-CE$_{224}$	72.0	52.8	85.3	77.1	**76.1**	58.4	70.4	61.5	**66.1**	**70.5**	99.2% (↓ 0.8%)
SUTrack-ToMe$_{224}$	71.4	52.0	85.2	**77.5**	**76.1**	59.2	69.4	61.7	65.9	**70.5**	99.0% (↓ 0.5%)
SUTrack-EViT$_{224}$	71.2	51.9	84.7	76.9	75.2	57.8	69.9	**62.3**	65.1	**70.5**	98.5% (↓ 1.5%)
SUTrack-DyViT$_{224}$	53.2	36.5	58.2	52.2	63.0	50.6	63.3	49.6	54.3	65.3	78.8% (↓ 21.2%)
UTPTrack-S$_{224}$ (ours)	**72.5**	**53.1**	**85.5**	77.1	75.7	**59.6**	**70.7**	61.7	65.9	70.1	**99.5%** (↓ 0.5%)
Retain 35.4% Tokens in Average (≈ 104 tokens, ↓ 64.6%)											
SUTrack-CE$_{224}$	70.5	51.8	84.8	75.9	75.5	57.8	69.3	61.4	65.4	**71.7**	98.3% (↓ 1.7%)
SUTrack-ToMe$_{224}$	67.9	48.6	80.0	71.6	70.9	53.8	63.6	58.0	61.7	67.3	92.5% (↓ 7.5%)
SUTrack-EViT$_{224}$	69.6	50.6	84.3	74.0	75.0	56.4	67.8	60.8	64.4	70.4	96.7% (↓ 3.3%)
SUTrack-DyViT$_{224}$	4.4	2.9	8.8	7.4	13.7	10.3	17.9	7.3	15.9	13.6	14.7% (↓ 85.3%)
UTPTrack-S$_{224}$ (ours)	**72.3**	**52.7**	**85.2**	**77.5**	**76.1**	**58.6**	**70.4**	**61.5**	**66.0**	70.3	**99.3%** (↓ 0.7%)

4.3. Ablation and Analysis

In this section, we ablate UTPTrack on both RGB-based and unified tracking, using OSTrack$_{256}$ and SUTrack$_{224}$ as respective baselines. We incrementally analyze the contribution of key components, including Search Region Candidate Elimination (CE), Dynamic Template Elimination (DTE), Static Template Elimination (STE), Token Type-Aware Pruning (TTA), and Text-Guided Pruning (TG). The results demonstrate that our framework effectively reduces token redundancy while preserving tracking accuracy, achieving an efficient performance trade-off.

Component Analysis in RGB-based Tracking. In Tab. 4, the baseline processes 384 tokens per layer at a cost of 34.5G MACs. With the stepwise addition of CE, DTE, and STE, tokens are reduced to 135 (65% pruned), and MACs to 23.8G (31% lower), with 98.9% performance maintained. Driven by a non-zero pruning budget and strong ST guidance, STE incurs a 0.7% drop, and TTA, which incorporates bounding-box priors to guide token pruning within ST, mitigates pruning errors and recovers 0.8%, reaching 99.7%. Notably, while DTE and STE (with TTA) further shrink tokens (321→271→135), performance is not monotonically decreasing: DTE yields +0.3% over CE and STE (with TTA) yields +0.1% over DTE, suggesting that pruning redundancy can improve generalization. Overall, the full framework delivers substantial redundancy and computation cuts with minimal loss.

Table 4. Ablation Study on **RGB-based Trackers**. Δ denotes the averaged performance change from the to row above.

#	Method	LaSOT	LaSOT$_{ext}$	TrkNet	GOT-10k	Avg Vis Tok	Cmp Vis Tok	MACs (G)	Avg Perf. (%)	Δ
1	Baseline	67.9	47.6	84.5	74.9	384	384	34.5	100.0	-
2	+ CE	67.4	46.6	83.7	75.5	291	217	27.0	99.3	↓ 0.7
3	+ DTE	68.4	46.6	83.8	75.3	271	176	25.4	99.6	↑ 0.3
4	+ STE	67.2	46.5	84.3	74.4	252	135	23.8	98.9	↓ 0.7
5	+ TTA	67.4	47.3	84.0	75.2	252	135	23.8	99.7	↑ 0.8

Component Analysis in Unified Tracking. Table 5 reports the ablation. The baseline processes 294 tokens per layer at 21.8G MACs. Mirroring the RGB case, CE, DTE, and STE (with TTA) reduces the average vision token to 188 and achieves 99.7% performance at 16.2G MACs. TG leverages language-based priors to highlight target-relevant semantics, further raising performance to 100.0%. These results demonstrate the robustness and generalizability of our pruning framework on unified tracking.

Table 5. Ablation Study on **Unified Trackers**. Δ denotes the average performance change from the row above.

#	Method	LaSOT	VOT-RGBD22	LasHeR	VisEvent	OTB99	Avg Vis Tok	Cmp Vis Tok	MACs (G)	Avg Perf. (%)	Δ
1	Baseline	73.7	75.5	59.9	63.3	68.4	294	294	21.8	100.0	-
2	+ CE	72.8	76.7	59.6	62.8	70.8	223	166	18.3	99.9	$\downarrow$ 0.1
3	+ DTE	72.8	76.3	60.3	62.5	70.8	206	128	17.3	99.7	$\downarrow$ 0.2
4	+ STE	72.3	76.7	60.1	61.4	70.1	188	90	16.2	99.3	$\downarrow$ 0.4
5	+ TTA	72.4	75.9	59.4	61.9	70.7	188	90	16.2	99.7	$\uparrow$ 0.4
6	+ TG	73.2	76.7	59.3	62.2	71.6	188	90	16.2	100.0	$\uparrow$ 0.3

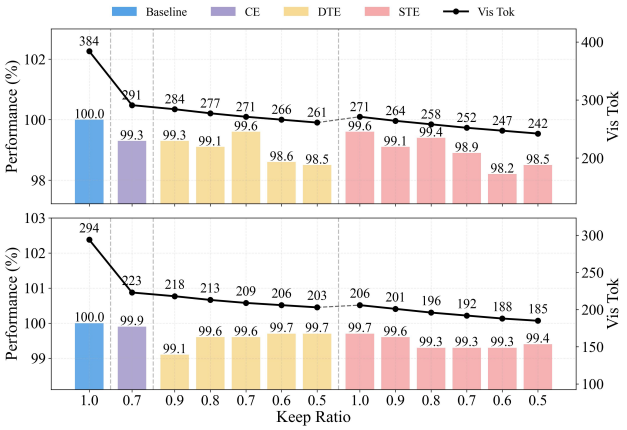


Figure 4. Ablation Study on Progressive Pruning. Performance and the number of vision tokens are reported as the keep ratio decreases and CE, DTE, and STE are progressively enabled for the RGB-based tracker (top) and unified tracker (bottom).

Progressive Pruning Analysis. As illustrated in Fig. 4, we perform a progressive pruning study on the RGB-based and unified baselines by sequentially removing tokens from SR, DT, and ST. This schedule steadily reduces the number of visual tokens while keeping performance closely aligned with the baseline for both trackers. Across most stages, accuracy stays within 1–2% of the original and only slightly drops under the most aggressive pruning ratio, indicating that a substantial fraction of SR, DT, and ST tokens is redundant and can be safely removed. Overall, we achieve large reductions in token count and computation with only negligible impact on tracking accuracy.

Pruning Location Analysis. As shown in Tab. 6, we systematically evaluate different layer-wise pruning configurations for the CE and DTE components in the unified tracker. Based on empirical results and a meticulous consideration of the trade-off between performance and efficiency, we select a configuration (#3) that prunes CE at layers [6, 12, 18] and DTE at layers [9, 15, 21], establishing a favorable balance between performance and efficiency. A corresponding study for the RGB-based tracker is provided in Appendix E.

Table 6. Ablation Study on CTEM Location for **Unified Trackers** with a 24-layer HiViT backbone.

#	CE Location	DTE Location	LaSOT	VOT-RGBD22	LasHeR	VisEvent	OTB99	Avg Vis Tok	Cmp Vis Tok	MACs (G)	Avg Perf. (%)
1	–	–	73.7	75.5	59.9	63.3	68.4	294	294	22.8	100.0
2	[6,12,18]	[6,12,18]	71.9	76.0	59.4	62.5	70.4	201	128	17.0	99.8
3	[6,12,18]	[9,15,21]	72.8	76.3	60.3	62.5	70.8	206	128	17.3	**100.6**
4	[6,12,18]	[3,9,15]	71.8	76.1	60.0	62.5	70.5	196	128	16.7	100.0
5	[3,9,15]	[6,12,18]	71.6	75.9	58.3	62.0	70.4	185	128	16.0	99.2
6	[9,15,21]	[6,12,18]	71.9	76.3	59.7	62.3	70.2	217	128	18.0	99.9

Table 7. Ablation study of selectively injecting language cues into different components for text-guided token pruning. Average performance is reported relative to the baseline of unified trackers.

#	Injection Target SR	DT	ST	LaSOT	VOT-RGBD22	LaHeR	VisEvent	OTB99	Avg Perf. (%)
1				72.4	75.9	59.4	61.9	70.7	99.7
2	✓			72.7	76.0	59.8	61.7	70.7	99.4
3		✓		73.2	76.7	59.3	62.2	71.6	**100.0**
4			✓	72.7	76.3	59.0	62.2	70.8	99.3
5	✓	✓		72.2	76.5	59.4	62.0	70.9	99.7
6	✓		✓	71.7	75.9	59.8	62.3	71.2	99.2
7		✓	✓	72.4	75.6	59.7	61.5	70.8	99.7
8	✓	✓	✓	72.9	76.0	59.3	62.3	70.5	99.5

Selective Injection of Language Cues. To further verify the effectiveness of TG, we conduct experiments by selectively injecting language cues into different target components, including SR, DT, and ST. As shown in Tab. 7, we evaluate all possible injection combinations. Experimental results show that injecting language cues into DT alone (#3) reaches 100.0%, matching the baseline. These suggest that DT is the most suitable locus for language-guided pruning, which we hypothesize stems from its higher foreground concentration. In contrast, injecting language into the SR or ST leads to noticeable performance drops.

5. Conclusion

This work introduced *UTPTrack*, a simple and unified token pruning framework that directly addresses the efficiency bottleneck in modern Transformer-based trackers. We presented the first method to jointly compress the search region, dynamic template, and static template, moving beyond the isolated pruning strategies of prior work. By leveraging attention-guided, token-type-aware mechanisms, UTPTrack effectively exploits cross-component redundancies to achieve significant computational savings. Its flexible design proves effective for both RGB tracking and more challenging multimodal and language-guided scenarios. Comprehensive evaluations on 10 benchmarks, we have shown that UTPTrack establishes a superior accuracy-efficiency balance, making it a powerful solution for developing high-performance yet practical visual trackers.

References

[1] Daniel Bolya, Cheng-Yang Fu, Xiaoliang Dai, Peizhao Zhang, Christoph Feichtenhofer, and Judy Hoffman. Token merging: Your vit but faster. *arXiv preprint arXiv:2210.09461*, 2022. 2, 5

[2] Boyu Chen, Peixia Li, Lei Bai, Lei Qiao, Qiuhong Shen, Bo Li, Weihao Gan, Wei Wu, and Wanli Ouyang. Backbone is all your need: A simplified architecture for visual object tracking. In *European conference on computer vision*, pages 375–392. Springer, 2022. 1, 2

[3] Fei Chen, Xiaodong Wang, Yunxiang Zhao, Shaohe Lv, and Xin Niu. Visual object tracking: A survey. *Computer Vision and Image Understanding*, 222:103508, 2022. 1

[4] Xin Chen, Bin Yan, Jiawen Zhu, Dong Wang, Xiaoyun Yang, and Huchuan Lu. Transformer tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8126–8135, 2021. 1

[5] Xin Chen, Ben Kang, Dong Wang, Dongdong Li, and Huchuan Lu. Efficient visual tracking via hierarchical cross-attention transformer. In *European Conference on Computer Vision*, pages 461–477. Springer, 2022. 1

[6] Xin Chen, Ben Kang, Wanting Geng, Jiawen Zhu, Yi Liu, Dong Wang, and Huchuan Lu. Sutrack: Towards simple and unified single object tracking. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2239–2247, 2025. 2, 4, 5

[7] Yutao Cui, Cheng Jiang, Limin Wang, and Gangshan Wu. Mixformer: End-to-end tracking with iterative mixed attention. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13608–13618, 2022. 1, 2

[8] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proceedings of the International Conference on Learning Representations*, 2021. 3

[9] Heng Fan, Liting Lin, Fan Yang, Peng Chu, Ge Deng, Sijia Yu, Hexin Bai, Yong Xu, Chunyuan Liao, and Haibin Ling. Lasot: A high-quality benchmark for large-scale single object tracking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5374–5383, 2019. 1, 5

[10] Heng Fan, Hexin Bai, Liting Lin, Fan Yang, Peng Chu, Ge Deng, Sijia Yu, Harshit, Mingzhen Huang, Juehuan Liu, et al. Lasot: A high-quality large-scale single object tracking benchmark. *International Journal of Computer Vision*, 129: 439–461, 2021. 1

[11] Yingqi Fan, Anhao Zhao, Jinlan Fu, Junlong Tong, Hui Su, Yijie Pan, Wei Zhang, and Xiaoyu Shen. Visipruner: Decoding discontinuous cross-modal dynamics for efficient multimodal llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 18896–18913, 2025. 2

[12] Kaijie He, Canlong Zhang, Sheng Xie, Zhixin Li, and Zhiwen Wang. Target-aware tracking with long-term context attention. In *Proceedings of the AAAI conference on artificial intelligence*, pages 773–780, 2023. 2

[13] Lingyi Hong, Shilin Yan, Renrui Zhang, Wanyun Li, Xinyu Zhou, Pinxue Guo, Kaixun Jiang, Yiting Chen, Jinglun Li, Zhaoyu Chen, et al. Onetracker: Unifying visual object tracking with foundation models and efficient tuning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19079–19091, 2024. 2

[14] Xiaojun Hou, Jiazheng Xing, Yijie Qian, Yaowei Guo, Shuo Xin, Junhao Chen, Kai Tang, Mengmeng Wang, Zhengkai Jiang, Liang Liu, et al. Sdstrack: Self-distillation symmetric adapter learning for multi-modal visual object tracking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26551–26561, 2024. 2

[15] Xiantao Hu, Bineng Zhong, Qihua Liang, Liangtao Shi, Zhiyi Mo, Ying Tai, and Jian Yang. Adaptive perception for unified visual multi-modal object tracking. *IEEE Transactions on Artificial Intelligence*, 2025. 2

[16] Lianghua Huang, Xin Zhao, and Kaiqi Huang. Got-10k: A large high-diversity benchmark for generic object tracking in the wild. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(5):1562–1577, 2019. 1, 5

[17] Zhenglun Kong, Peiyan Dong, Xiaolong Ma, Xin Meng, Wei Niu, Mengshu Sun, Xuan Shen, Geng Yuan, Bin Ren, Hao Tang, et al. Spvit: Enabling faster vision transformers via latency-aware soft token pruning. In *European conference on computer vision*, pages 620–640. Springer, 2022. 2

[18] Matej Kristan, Aleš Leonardis, Jiří Matas, Michael Felsberg, Roman Pflugfelder, Joni-Kristian Kämäräinen, Hyung Jin Chang, Martin Danelljan, Luka Čehovin Zajc, Alan Lukežič, et al. The tenth visual object tracking vot2022 challenge results. In *European Conference on Computer Vision*, pages 431–460. Springer, 2022. 1

[19] Jin-Peng Lan, Zhi-Qi Cheng, Jun-Yan He, Chenyang Li, Bin Luo, Xu Bao, Wangmeng Xiang, Yifeng Geng, and Xuansong Xie. Procontext: Exploring progressive context transformer for tracking. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2023. 2, 3

[20] Jaeyeon Lee and Dong-Wan Choi. Lossless token merging even without fine-tuning in vision transformers. *arXiv preprint arXiv:2505.15160*, 2025. 2

[21] Chenglong Li, Xinyan Liang, Yijuan Lu, Nan Zhao, and Jin Tang. Rgb-t object tracking: Benchmark and baseline. *Pattern Recognition*, 96:106977, 2019. 1

[22] Chenglong Li, Wanlin Xue, Yaqing Jia, Zhichen Qu, Bin Luo, Jin Tang, and Dengdi Sun. Lasher: A large-scale high-diversity benchmark for rgbt tracking. *IEEE Transactions on Image Processing*, 31:392–404, 2021. 5, 1

[23] Zhenyang Li, Ran Tao, Efstratios Gavves, Cees GM Snoek, and Arnold WM Smeulders. Tracking by natural language specification. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6495–6503, 2017. 1

[24] Youwei Liang, Chongjian Ge, Zhan Tong, Yibing Song, Jue Wang, and Pengtao Xie. Not all patches are what you

need: Expediting vision transformers via token reorganizations. *arXiv preprint arXiv:2202.07800*, 2022. 2, 5

[25] Liting Lin, Heng Fan, Zhipeng Zhang, Yong Xu, and Haibin Ling. Swintrack: A simple and strong baseline for transformer tracking. *Advances in Neural Information Processing Systems*, 35:16743–16754, 2022. 1

[26] Liting Lin, Heng Fan, Zhipeng Zhang, Yaowei Wang, Yong Xu, and Haibin Ling. Tracking meets lora: Faster training, larger model, stronger performance. In *European Conference on Computer Vision*, pages 300–318. Springer, 2024. 4

[27] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Proceedings of the European Conference on Computer Vision*, pages 740–755, 2014. 5

[28] Chang Liu, Ziqi Guan, Simiao Lai, Yang Liu, Huchuan Lu, and Dong Wang. Emtrack: Efficient multimodal object tracking. *IEEE Transactions on Circuits and Systems for Video Technology*, 2024. 2

[29] Junzhu Mao, Yang Shen, Jinyang Guo, Yazhou Yao, Xiansheng Hua, and Hengtao Shen. Prune and merge: Efficient token compression for vision transformer with spatial information preserved. *IEEE Transactions on Multimedia*, 2025. 2

[30] Matthias Muller, Adel Bibi, Silvio Giancola, Salman Alsubaihi, and Bernard Ghanem. Trackingnet: A large-scale dataset and benchmark for object tracking in the wild. In *Proceedings of the European Conference on Computer Vision*, pages 300–317, 2018. 5, 1

[31] Liang Peng, Junyuan Gao, Xinran Liu, Weihong Li, Shaohua Dong, Zhipeng Zhang, Heng Fan, and Libo Zhang. Vasttrack: Vast category visual object tracking. *Advances in Neural Information Processing Systems*, 37:130797–130818, 2024. 5

[32] Shuai Peng, Di Fu, Baole Wei, Yong Cao, Liangcai Gao, and Zhi Tang. Vote&mix: Plug-and-play token reduction for efficient vision transformer. *arXiv preprint arXiv:2408.17062*, 2024. 2

[33] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021. 3, 5

[34] Yongming Rao, Wenliang Zhao, Benlin Liu, Jiwen Lu, Jie Zhou, and Cho-Jui Hsieh. Dynamicvit: Efficient vision transformers with dynamic token sparsification. *Advances in neural information processing systems*, 34:13937–13949, 2021. 2, 5

[35] Xiao Wang, Xiujun Shu, Zhipeng Zhang, Bo Jiang, Yaowei Wang, Yonghong Tian, and Feng Wu. Towards more flexible and accurate object tracking with natural language: Algorithms and benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13763–13773, 2021. 5, 1

[36] Xiao Wang, Jianing Li, Lin Zhu, Zhipeng Zhang, Zhe Chen, Xin Li, Yaowei Wang, Yonghong Tian, and Feng Wu. Visevent: Reliable object tracking via collaboration of frame and event flows. *IEEE Transactions on Cybernetics*, 54(3): 1997–2010, 2023. 5, 1

[37] Yuanhui Wang, Ben Ye, and Zhanchuan Cai. Dynamic template updating using spatial-temporal information in siamese trackers. *IEEE Transactions on Multimedia*, 26:2006–2015, 2023. 1

[38] Yuanhui Wang, Ben Ye, Zhanchuan Cai, and Hao Wu. A transformer-based tracker integrating motion and representation information. *IEEE Transactions on Multimedia*, pages 1–10, 2026. 1

[39] Hao Wu, Junlong Tong, Xudong Wang, Yang Tan, Changyu Zeng, Anastasia Antsiferova, and Xiaoyu Shen. From data to model: A survey of the compression lifecycle in mllms. 2026. 2

[40] Zongwei Wu, Jilai Zheng, Xiangxuan Ren, Florin-Alexandru Vasluianu, Chao Ma, Danda Pani Paudel, Luc Van Gool, and Radu Timofte. Single-model and any-modality for video object tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19156–19166, 2024. 2

[41] Fei Xie, Chunyu Wang, Guangting Wang, Yue Cao, Wankou Yang, and Wenjun Zeng. Correlation-aware deep tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8751–8760, 2022. 1

[42] Yifan Xu, Zhijie Zhang, Mengdan Zhang, Kekai Sheng, Ke Li, Weiming Dong, Liqing Zhang, Changsheng Xu, and Xing Sun. Evo-vit: Slow-fast token evolution for dynamic vision transformer. In *Proceedings of the AAAI conference on artificial intelligence*, pages 2964–2972, 2022. 2

[43] Bin Yan, Houwen Peng, Jianlong Fu, Dong Wang, and Huchuan Lu. Learning spatio-temporal transformer for visual tracking. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10448–10457, 2021. 1, 5

[44] Song Yan, Jinyu Yang, Jani Käpylä, Feng Zheng, Aleš Leonardis, and Joni-Kristian Kämäräinen. Depthtrack: Unveiling the power of rgbd tracking. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10725–10733, 2021. 5

[45] Botao Ye, Hong Chang, Bingpeng Ma, et al. Joint feature learning and relation modeling for tracking: A one-stream framework. In *ECCV*, pages 341–357. Springer, 2022. 1, 2, 3, 4, 5, 6

[46] Shuo Zhang, Dan Zhang, and Qi Zou. Atptrack: Visual tracking with alternating token pruning of dynamic templates and search region. *Neurocomputing*, 625:129534, 2025. 2

[47] Jiawen Zhu, Simiao Lai, Xin Chen, Dong Wang, and Huchuan Lu. Visual prompt multi-modal tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9516–9526, 2023. 2

[48] Jiawen Zhu, Huayi Tang, Xin Chen, Xinying Wang, Dong Wang, and Huchuan Lu. Two-stream beats one-stream: asymmetric siamese network for efficient visual tracking. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 10959–10967, 2025. 1